



Programmable Wearables for Human Perception

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Resumo

To protect Vulnerable Road Users from urban blind spots, we introduce a mobile, end-to-end architecture featuring:

- **Edge-to-cloud:** Real-time context awareness for safety events.
- **Smartphone hub:** Orchestrates multiple connected wearables
- **Synchronization:** Aligns independent multimodal streams into a unified timeline.

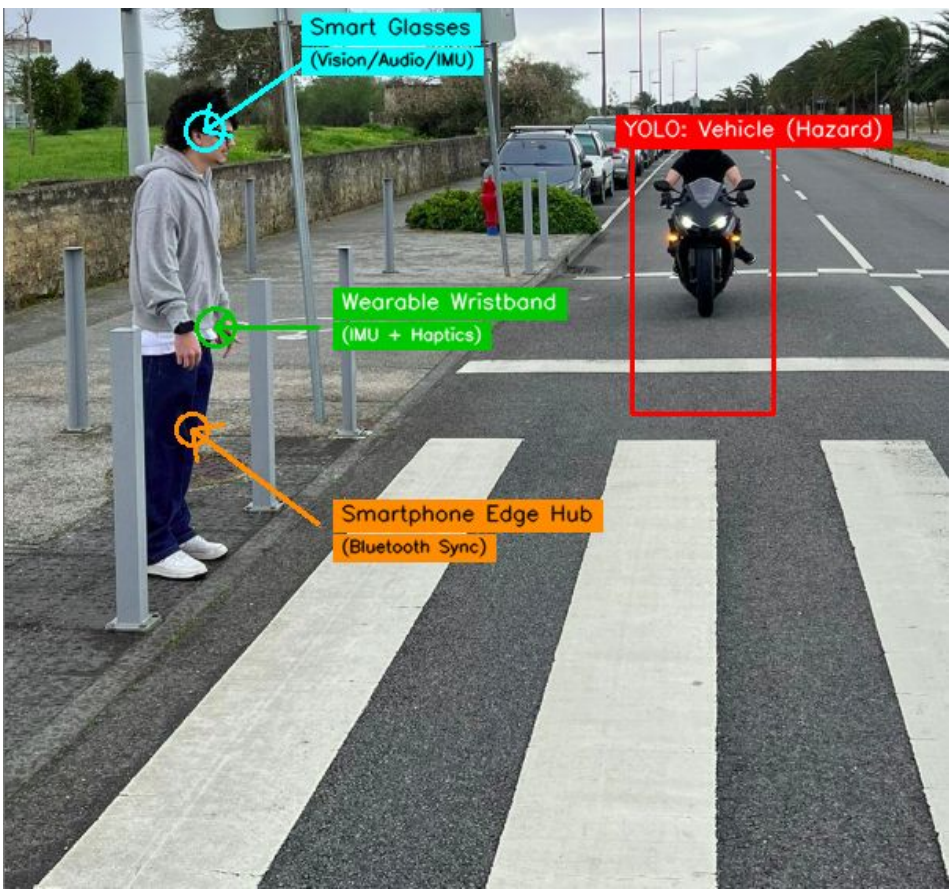


Fig 1- Project Context

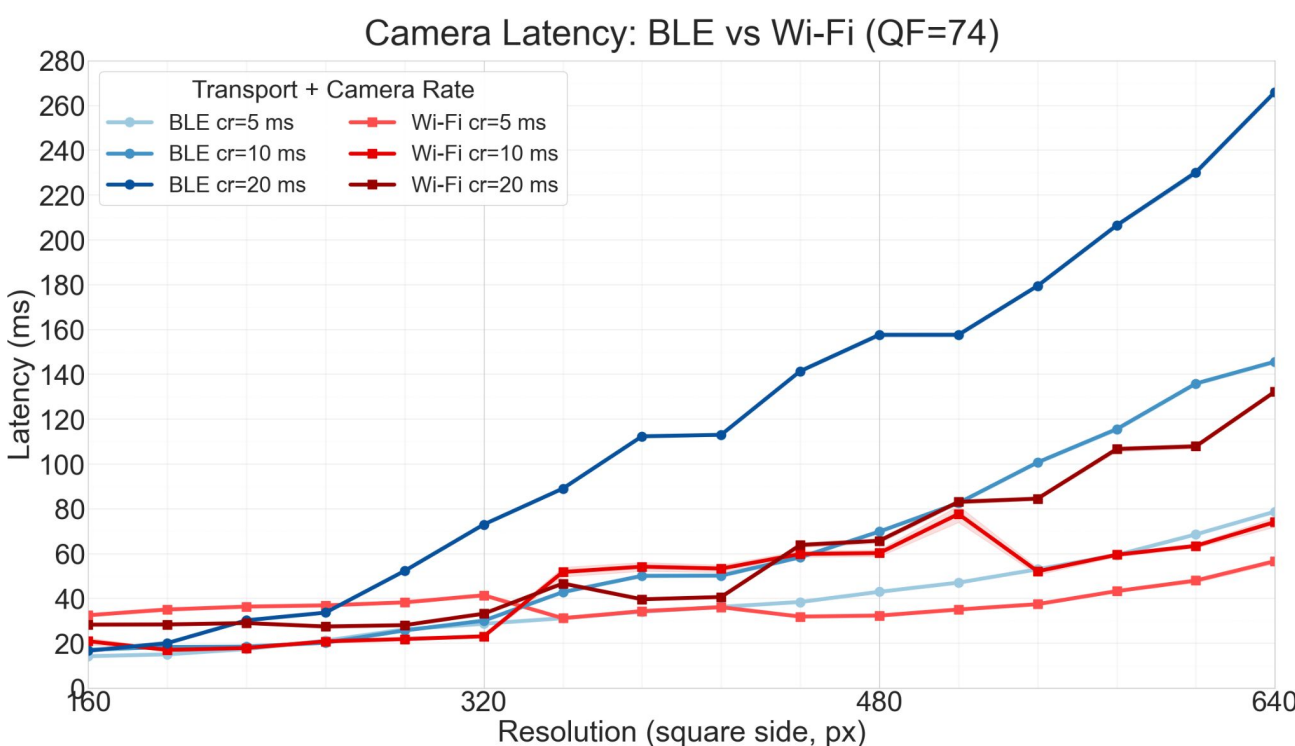


Fig 3- Camera Latency results WIFI/BLE

Results:

- **High-Speed Multimedia Streaming:** The dual channel architecture handles continuous HVGA video alongside IMU telemetry
- **Edge Transport Latency:** Achieved a mean WIFI/UDP transport latency of 33.5ms(±13.8ms).
- **Cloud Inference Speed:** Backend processing is in the range of ~45-60 ms per image in YOLO11n
- **Temporal Precision:** Hardware clock synchronization fuses footsteps, vehicle sightings, and speech into the Fusion Engine

Methods

The architecture is divided into two primary tiers:

- **Distributed Pipeline:** Processing is splitted across wearable sensors (capture), the smartphone (sync), and the cloud(inference)
- **The android Edge hub:** Maintains concurrent BLE wearable connection while also reassembling video transmitted packets via UDP.
- **The Cloud Server:** Processes video stream via YOLO11n, DepthAnything V2, and sherpa.

Conclusion

- **Architectural Success:** Proves a standard smartphone can orchestrate heterogeneous wearables into a cohesive, real-time safety network.
- **Direct VRU Protection:** Bypasses infrastructure by giving users continuous, localizes context awareness.

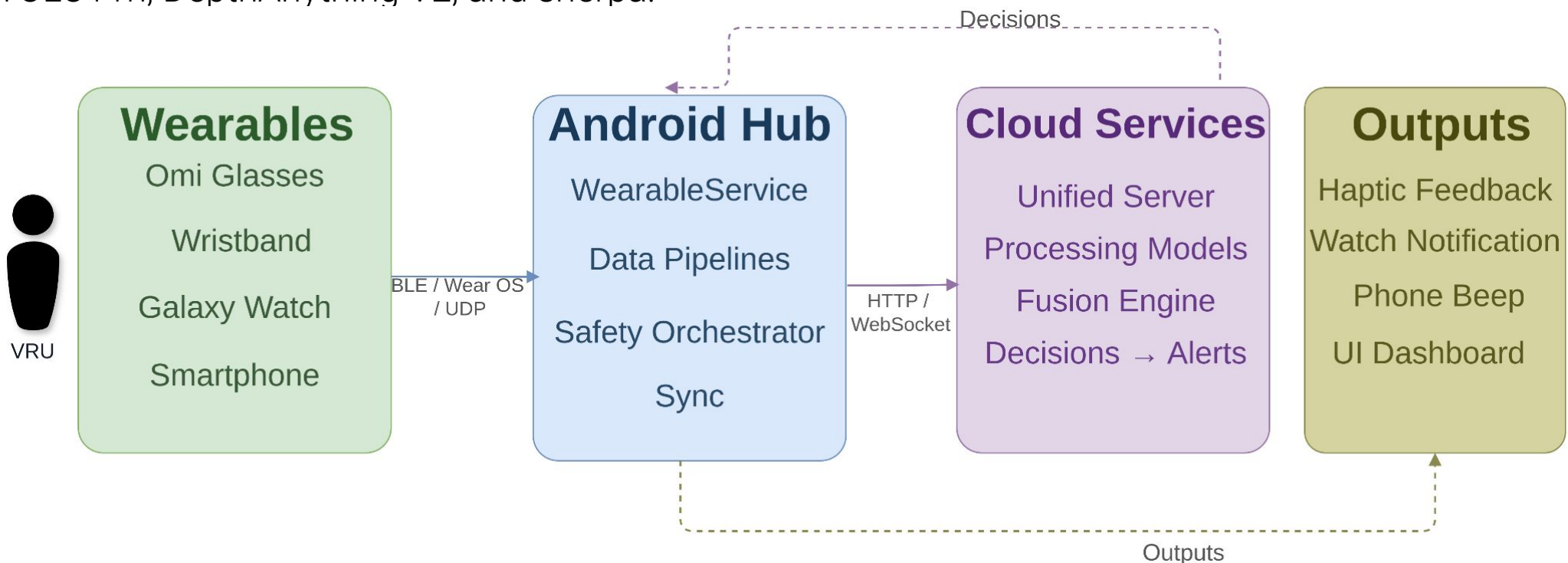


Fig 2- System Architecture .